

End-of-Course Project

Learning-based Registration of the Cervical Phantom without Prescribed Fiducial Points

Medical Robotics Course

Academic Year 2025/2026

1 Context and goal

The laboratory owns a physical–digital phantom of the cervical district (3D-printed, muscle-focused; the pink skin and outer shell are removed so that the exposed internal anatomy is accessible). A simplified digital twin of the phantom is available as a triangular mesh (≈ 173 k vertices, ≈ 211 k faces) and acts as the ground-truth model.

The **current registration method** is correspondence-based: four points whose coordinates are known in the phantom frame (origin at the bottom-centre, right-handed, millimetres) are touched in a prescribed order with a probe tracked by an **NDI Polaris Vega XT** optical tracker; the rigid transformation between the Polaris frame and the phantom frame is then obtained in closed form (the SVD / Procrustes solution from known correspondences [1]).

Goal of the project. Keep the same hardware (Polaris Vega XT and the tracked probe) but **remove the dependence on known points and on the prescribed touching order**. Instead, the operator acquires an *unordered, partial* point set from the exposed phantom surface, and a learning-based method estimates the same Polaris→phantom rigid transformation by aligning that point set to the digital twin. The existing four-point method is the **baseline** that every variant must be compared against, using the same target registration error (TRE).

2 Acquisition modes and reference data

Two acquisition modes are studied and explicitly compared (this is a required experimental axis, not an implementation detail):

- **Sweep:** the operator drags the tracked probe over the exposed internal anatomy (muscle bundles, cavity walls) and the rim, producing a dense, connected, partial point cloud.
- **Sparse:** the operator touches 20–50 arbitrary (*not* prescribed, *not* ordered) points on the reachable surface.

Because the digital twin *is* the ground-truth model, training and evaluation data can be generated essentially for free, with exact ground truth:

1. sample points on the digital-twin surface;
2. apply a known random rigid transformation spanning the realistic Polaris working volume and angular range;
3. add Gaussian noise at the Polaris-tip level ($\sigma \approx 0.25$ – 0.35 mm RMS, consistent with a Vega XT probe after pivot calibration);
4. crop to the probe-reachable regions to emulate partial overlap — a dense connected patch for the *sweep* mode, 20–50 scattered points for the *sparse* mode.

A small set of *real* acquisitions with the Polaris should also be collected on the physical phantom for a final reality check, but the bulk of training/validation is synthetic.

3 Project variants

The project is offered as **three variants**; a group implements one. All three estimate the Polaris→phantom transform and are scored against the four-point baseline on both acquisition modes.

ID	Method	Role w.r.t. current method	Best-suited mode
V1	PointNetLK [3] (correspondence-free)	Removes correspondences entirely; learned counterpart of ICP	Robust on <i>sparse</i> ; works on <i>sweep</i>
V2	Deep Closest Point [4] (correspondence-learned)	Keeps the SVD/Procrustes head of the current method, but <i>learns</i> the correspondences instead of prescribing them	Both; the most direct conceptual upgrade
V3	FCGF [5] descriptors + RANSAC, then ICP refine	Replaces hand-picked fiducials with learned 3D features; coarse pose refined by classical ICP	Strong on <i>sweep</i> ; weak on <i>sparse</i>

3.1 V1 — Correspondence-free (PointNetLK)

Idea: PointNet acts as a learnable global descriptor; an inverse-compositional Lucas–Kanade iteration, unrolled as a trainable network, recovers the rigid transform with no point matches [3].

Why it fits this phantom: with sparse, scattered probe touches there are no reliable local correspondences, so a correspondence-free formulation is the natural choice for that mode.

Expected behaviour: most robust in the *sparse* mode; sensitive to large initial misalignment (study the basin of attraction).

Compare: TRE and convergence vs the four-point baseline, for sweep and sparse separately.

3.2 V2 — Correspondence-learned (Deep Closest Point)

Idea: DCP learns a point embedding and an attention-based soft assignment, then recovers the transform with a differentiable SVD layer [4]. The SVD head solves exactly the same Procrustes problem as [1] — the only thing that changes is that the correspondences are *learned* rather than prescribed by the touch order.

Why it fits this phantom: it is the minimal conceptual change from the existing method and therefore the clearest teaching path; you can reuse your understanding of the closed-form solver.

Expected behaviour: works in both modes; the natural primary recommendation.

Compare: TRE vs the four-point baseline and, as a sanity check, vs the closed-form solution on the original four points.

3.3 V3 — Learned descriptors + RANSAC + ICP (FCGF)

Idea: a fully convolutional network produces dense 3D feature descriptors; nearest-neighbour matching in feature space plus RANSAC yields a robust coarse pose, which classical ICP then refines [5,6,7].

Why it fits this phantom: the exposed cervical anatomy and the asymmetric cradle/base

geometry are sufficiently distinctive for descriptor matching *when a dense sweep is available*; this mirrors the learned-coarse / ICP-fine pipeline used in real surgical navigation [6,7].

Expected behaviour: strong in the *sweep* mode, expected to degrade in the *sparse* mode (insufficient local context for descriptors) — documenting this contrast is part of the deliverable.

Compare: learned-descriptor + RANSAC + ICP vs the four-point baseline, reporting both coarse and refined error.

4 Evaluation protocol

- **Primary metric — TRE.** Use the same target registration error definition as [9], evaluated at clinically meaningful targets. At least one target must be placed near the **spinal cord**, since that is the structure that must not be damaged and where accuracy matters most.
- **Secondary metrics.** Rotation error (deg), translation error (mm), and, for the sweep/sparse comparison, registration success rate and the size of the initial-misalignment basin within which the method still converges.
- **Required comparison matrix.** Every variant reports a $2 \times K$ table: {sweep, sparse} $\times$ {learned method, four-point baseline (+ FPFH baseline for V3)}, on synthetic data with exact ground truth, plus a small real-Polaris confirmation set.
- **Honest reporting.** The four-point method uses only four points but exact correspondences; the learned methods use far more surface information but no correspondences. The interesting question — which the report must answer — is whether removing the prescribed-point requirement can match or beat the current TRE, and under which acquisition mode.

5 Practical and organisational notes

- **Starting point.** Use the authors’ reference implementations [3,4,5]; the expected contribution is a clear, reproducible adaptation to this phantom and acquisition setup, not a new method.
- **Coordinate frames.** The method must output the Polaris→phantom rigid transform (the ${}^t\mathbf{T}_s$ of the thesis). The subsequent right- to left-handed Unity conversion (y/z swap, /1000) is fixed and reused unchanged.
- **Probe calibration.** Assume the probe tip is pivot- calibrated as in the current pipeline; the acquired quantity is the tip position in the Polaris frame.
- **Deliverables.** Reproducible code, the data-generation script, a short technical report (method, the sweep/sparse comparison, honest comparison with the four-point baseline, failure cases), and a final presentation.

6 References

- [1] K. S. Arun, T. S. Huang, S. D. Blostein, “Least-Squares Fitting of Two 3-D Point Sets,” *IEEE TPAMI*, PAMI-9(5), 1987.
- [2] P. J. Besl, N. D. McKay, “A Method for Registration of 3-D Shapes,” *IEEE TPAMI*, 14(2), 1992.
- [3] Y. Aoki, H. Goforth, R. A. Srivatsan, S. Lucey, “PointNetLK: Robust & Efficient Point Cloud Registration Using PointNet,” *CVPR*, 2019. arXiv:1903.05711. Code: <https://github.com/hmgoforth/PointNetLK>
- [4] Y. Wang, J. M. Solomon, “Deep Closest Point: Learning Representations for Point Cloud Registration,” *ICCV*, 2019. arXiv:1905.03304. Code: <https://github.com/WangYueFt/dcp>
- [5] C. Choy, J. Park, V. Koltun, “Fully Convolutional Geometric Features,” *ICCV*, 2019. Code: <https://github.com/chrischoy/FCGF>

- [6] T. Liu *et al.*, “A Spatial Registration Method Based on Point Cloud and Deep Learning for Augmented Reality Neurosurgical Navigation,” *Int. J. Medical Robotics and Computer Assisted Surgery*, 2024.
 - [7] Deep Learning-Based Fine-Tuning Approach of Coarse Registration for Ear–Nose–Throat (ENT) Surgical Navigation Systems, *Sensors*, 2024.
 - [8] M. Weber, D. Wild, J. Kleesiek, J. Egger, C. Gsaxner, “Deep Learning-based Point Cloud Registration for Augmented Reality-guided Surgery,” arXiv:2405.03314, 2024 (cautionary benchmark: a well-tuned classical pipeline can still be competitive).
 - [9] A Master Thesis provided by the instructor.
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